

Diego Barrio Hortigüela

Senior Backend Engineer · Java, Spring & Applied AI

Cloud-native systems · Distributed architectures · RAG and agentic workflows

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About

Senior Backend Engineer with over 15 years of experience designing and building distributed systems, payment platforms, and cloud-native applications. Specialized in Java, Spring Boot, event-driven architectures, and microservices, with hands-on experience operating containerized workloads on Kubernetes. Experienced in fintech, banking, and mobility domains, delivering scalable and resilient backend systems. Currently focused on Platform Engineering, Observability, and AI-powered development workflows, exploring agentic systems, LLM integrations, and modern software delivery practices.

Core Competencies

Backend Engineering

Java · Spring Boot · Spring Security · Spring Data JPA · Hibernate · REST APIs · TDD

Software Architecture

Microservices · DDD · Hexagonal Architecture · Clean Architecture · Event-Driven · API Design

Platform Engineering

Docker · Kubernetes · GitLab CI/CD · Jenkins · AWS · Azure

Observability

Grafana · Prometheus · Loki · OpenTelemetry · Sentry · Splunk

Data & Messaging

Oracle · SQL Server · PostgreSQL · Apache Kafka

AI Engineering

LLM Integration · Agentic Workflows · Tool Calling · MCP · Ollama · Claude Code · OpenAI APIs · Gemini · RAG

Featured Projects

Akademia

RAG-Powered Learning Platform

Web app to study and prepare for exams. The core of the system is the RAG generation pipeline: it turns the resources a user uploads into tests and questions automatically.

- RAG generation pipeline that produces tests and questions from user-supplied material.
- Embeddings through the OpenAI embeddings API via OpenRouter; Gemini Embedding 2 support in progress.
- Chunking, indexing, and retrieval tuned so generated questions stay grounded in the source.
- Spaced-repetition flashcards and self-assessment.
- Integrated Stripe store to purchase study materials.

Technologies: Java, Spring Boot, PostgreSQL, RAG, OpenAI Embeddings, OpenRouter, Stripe

TokenMeter

Tokenization & LLM Cost Exploration

An open project exploring tokenization and the real cost of LLMs in code generation, across both agentic loops and more direct single-shot flows.

- Repository token analysis engine.
- Cost comparison of agentic versus direct code-generation flows.
- Token counting with the jtokkit o200k_base encoder rather than a character-based approximation.
- Deployed behind Cloudflare.

Technologies: Java, Spring Boot, Docker, Cloudflare, LLM APIs

Forma

Personal Fitness Operating System

A personal health and fitness planning system that connects body composition, training, nutrition and shopping cost into one weekly decision loop.

- Hexagonal backend with a framework-free domain: JDBC and Flyway over PostgreSQL, no ORM, decided and recorded in an ADR.
- LLM-generated nutrition plans are fenced by a validated contract: a plan may only reference food identifiers the API publishes, and anything the model invents is rejected on import.
- Withings OAuth integration for automatic body-measurement sync.
- Weekly insights service that turns raw metrics into small, actionable adjustments.
- React 19 front end with Vitest unit tests and Playwright end-to-end coverage, all gated in CI.
- Spec-driven delivery: 13 architecture decision records and per-story specs kept in the repository.

Technologies: Java 21, Spring Boot 3, PostgreSQL, Flyway, React 19, TypeScript, Vite, Playwright, Docker Compose

Backend to the Future

Engineering Blog & Technical Platform

Personal platform where I publish projects, technical articles, and engineering experiments.

- Backend development.
- Software architecture.
- Cloud-native systems.
- Kubernetes.
- AI engineering.
- Developer productivity.

Local AI Lab

Self-Hosted AI Infrastructure

Personal environment dedicated to evaluating and operating local Large Language Models.

- Local LLM deployment.
- Ollama and LM Studio.
- Models tested: Qwen 3.5, Gemma 3, Llama 3.1, DeepSeek R1, and Mistral Nemo.
- Agentic software development workflows.
- MCP integrations.
- Autonomous coding assistants.

Hardware: NVIDIA GeForce RTX 3060 (12 GB VRAM)

Professional Experience

Senior Backend Engineer

Mindden (for Europcar Mobility Group) · Mar 2024 – Present

- Designed and implemented billing and payment APIs using Spring Boot, microservices, AWS, and Kubernetes.
- Integrated Apache Kafka to support event-driven architectures and asynchronous processing.
- Promoted Domain-Driven Design, Hexagonal Architecture, and TDD practices.
- Improved CI/CD pipelines using GitLab.
- Contributed to cloud-native platform initiatives and observability practices.

Technologies: Java, Spring Boot, Kafka, Kubernetes, AWS, GitLab CI/CD, Oracle

Senior Analyst

Accenture · Jul 2022 – Mar 2024

- Developed and maintained banking systems using Spring Boot and Hibernate.
- Built microservices focused on scalability and reliability.
- Managed batch processing workflows using IBM Tivoli Workload Scheduler.
- Optimized complex Oracle and SQL Server queries.

Technologies: Java, Spring Boot, Oracle, SQL Server, IBM Tivoli

Senior Specialist

SABIS · Jun 2014 – Jun 2022

- Developed and maintained enterprise platforms used across multiple departments.
- Built applications using Spring MVC, WebFlow, JPA, and Hibernate.
- Participated in integrations involving .NET and SharePoint solutions.

Technologies: Java, Spring MVC, Hibernate, Oracle, SharePoint

Analyst

Banco Sabadell · Nov 2012 – Jun 2014

- Developed internal business applications using the Spring ecosystem.
- Participated in the design and implementation of web-based enterprise solutions.

Technologies: Java, Spring, Thymeleaf, Oracle

Analyst & Developer

Caja Mediterráneo (CAM) · Mar 2007 – Nov 2012

- Developed and maintained online banking applications.
- Participated in ATM services and core banking migration projects.

Technologies: Java EE, Oracle

Web Developer

HDV · Dec 2005 – Mar 2007

- Developed dynamic web applications and interactive user interfaces.

Technologies: PHP, MySQL, JavaScript

Education

Master's Degree in Web Applications and Services Development

Universidad de Alicante · 2010 – 2011

Certificate in Java Enterprise Development

Universidad de Alicante · 2009 – 2010

Computer Engineering Degree

Universidad de Alicante · 1998 – 2005

Equivalent to MSc level under the Bologna Process.

Community & Personal Projects

- Creator of Backend to the Future, a platform focused on software architecture and backend engineering.
- Creator of Akademia, a learning platform whose core is a RAG pipeline that generates exams from user-supplied material.
- Creator of TokenMeter, an open exploration of tokenization and LLM cost in code generation.
- Creator of Forma, a personal fitness operating system where LLM-generated plans are constrained by a validated domain contract.
- Active experimentation with local LLMs, AI agents, and autonomous development workflows.
- Former mentor for Computer Science students through Fundación Banco Sabadell.

Learning Focus

Platform Engineering · Advanced Kubernetes · Observability · SRE Principles ·
AI Agents & Autonomous Workflows · Rust

Languages

Language	Level
Spanish	Native
English	Full Professional Proficiency (C1)